CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

IV Semester of M. Sc. (AOC) Examination May 2018

OC 831 Selected Topics in Organic Chemistry

Date: 03/05/2018 Day: Thursday Time: 10.00 am To 10.30 am Maximum Marks: 20

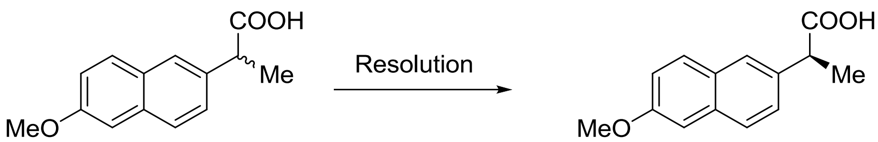
MCQ

Important Instructions:

- Tick the correct answer and it should be written in question paper itself.
- Use of non-programmable calculator is allowed.

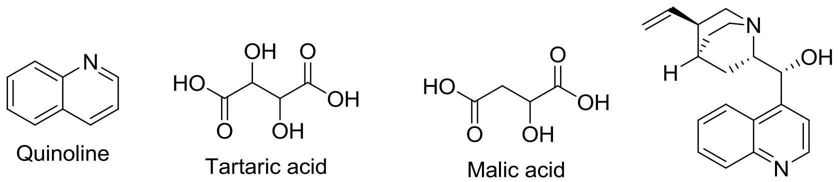
Q - I Choose the correct answer for the following questions. [20]

1 Naproxen has been synthesized as a racemate then resolved by crystallization of diastereomeric salts.



Naproxen

Which of the following compounds would be suitable for preparing the necessary diastereomeric salts?



Quinoline Tartaric acid Malic acid Cinchonidine

a) Quinoline

b) Tartaric acid

c) Malic acid

d) Cinchonidine

2 A reaction in which the reactant cannot exist in stereo isomeric forms and it produces predominantly or exclusively one stereo isomeric form of the product is known as ....

a) stereo selective reaction   b) stereospecific reaction

c) epimerization reaction   d) mesomeric reaction

3 In a bifunctional compound, if reagent reacts with one functional group preferentially, even though the other functional group is apparently susceptible to the reaction conditions, then the reaction is called ....

a) Stereoselective   b) Chemoselective

c) Stereospecific   d) Regioselective

4 The reaction of BH_3 with 1-butene in presence of H_2O_2 leads to formation of ....

a) Racemic mixture

b) (S)-2-Hydroxybutane major and (R)-2-Hydroxybutane minor

c) (S)-2-Hydroxybutane minor and (R)-2-Hydroxybutane major

d) Only (S)-2-Hydroxybutane

5 For a substance to generate heat when irradiated under microwave irradiation, it must ___________________.

a) have higher refractive index   b) be soluble in solvent

c) have dipole moment   d) be non polar

6 If acetone and ethanol heated under same microwave irradiation power and for the same period of time, then the final temperature will much higher in ethanol compare to acetone, which is due to

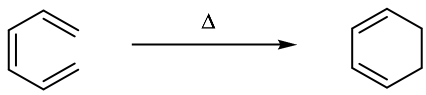
a) ethanol has higher dielectric constant than acetone

b) ethanol has higher tan δ value than acetone

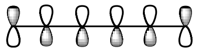
c) acetone has higher tan δ value than ethanol

d) acetone has higher dielectric constant than ethanol

7 According to Frontier Molecular Orbital (FMO) theory, the Highest Occupied Molecular Orbital (HOMO) of hexatriene in the following reaction is ....



a)



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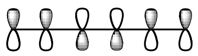
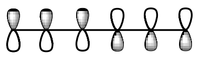
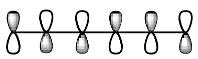


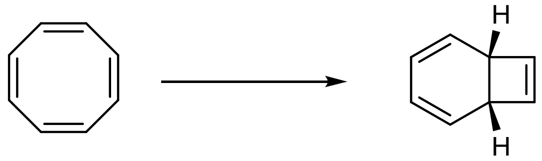
c)



d)

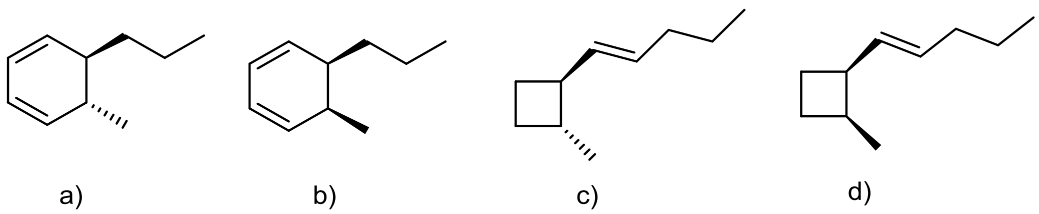


8 The most appropriate mode of cyclisation in the following transformation is ……….



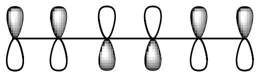
a) Con-rotatory in Photochemical; and Dis-rotatory in Thermal
b) Con-rotatory in Thermal; and Dis-rotatory in Photochemical
c) Con-rotatory in Photochemical; and Con-rotatory in Thermal
d) Dis-rotatory in Photochemical; and Dis-rotatory in Thermal

9 The major product formed by thermal induced reaction of (2E,4Z,6E)-decatriene is ……….



10 Which of the following molecular orbital(s) would undergo disrotatory ring closure?

a)



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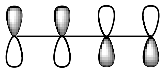
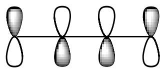
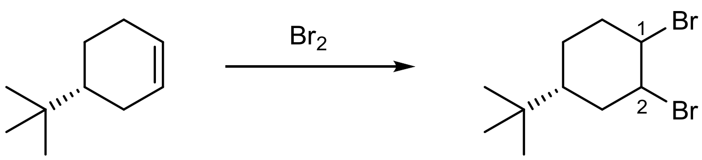


c)



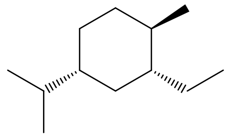
d) none of above

11 In the reaction given below, the orientation of two bromine substituents in the product is ……….



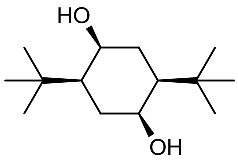
a) Equatorial at both C-1 and C-2
b) Equatorial at C-1 and Axial at C-2
c) Axial at C-1 and Equatorial at C-2
d) Axial at both C-1 and C-2

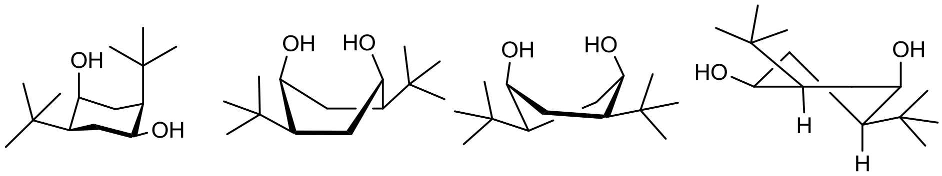
12 In the lowest energy conformation of the compound below, how many alkyl substituents are axial?



a) 2 b) 0 c) 3 d) 1

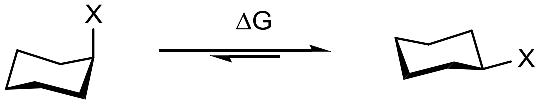
13 The most stable conformation for the following compound is ……….





a) b) c) d)

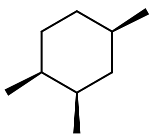
14 The correct order of magnitude of “A values” for the given substituents in cyclohexane derivative is ……….

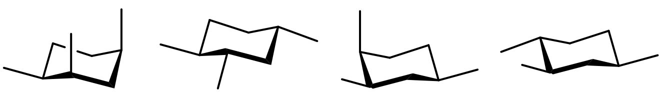


X = CH_3
X = CN
X = Ph

a) Ph > CN > Me
b) Me > Ph > CN
c) CN > Me > Ph
d) Ph > Me > CN

15 Among the structure given below, the most stable conformation for the following compound is ……….

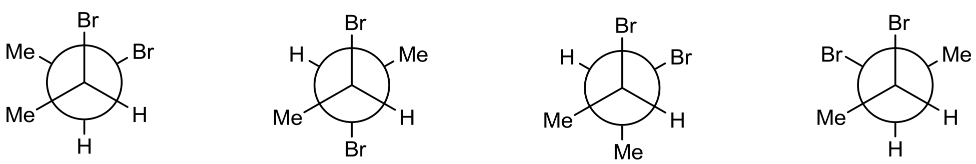




a) b) c) d)

16 The gauche interaction values for Me/Me, Me/Br and Br/Br are 3.3, 0.8 and 3.0 Kilojoule/Mole, respectively. Among the following, the most stable conformation of 2,3-dibromobutane is ……….

a) b) c) d)



17 The chair conformation of cyclohexane contains

a) Six equatorial hydrogens and Six axial hydrogens

b) All twelve equatorial hydrogens

c) All twelve axial hydrogens

d) Three equatorial hydrogens and Three axial hydrogens

18 Which of the following correctly lists the conformations of cyclohexane in order of decreasing energy?

a) chair > twist-boat > boat > half – chair

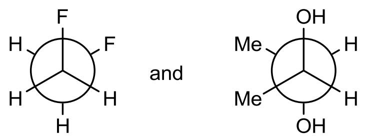
b) chair > boat > twist – boat > half – chair

c) half – chair > boat > twist – boat > chair

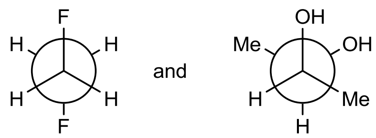
d) chair > twist – boat > half – chair > boat

19 The most stable conformation of 1,2-difluoroethane and dl-2,3-butandiol are ……….

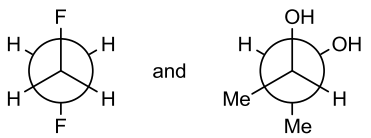
a)



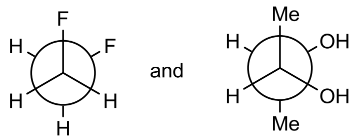
b)



c)



d)



20 In the lowest energy chair conformation of cis-1,4-dimethyl cyclohexane, how many axial position are occupied by hydrogen atoms?

a) 4 b) 6 c) 3 d) 5

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OC 831 Selected Topics in Organic Chemistry

Date: 03/05/2018 Day: Thursday Time: 10.30 am To 01.00 pm Maximum Marks: 50

Important Instructions:

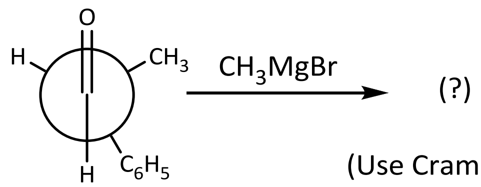
- **Section I and II must be attempted in Two Separate ANSWER SHEET.**
- Make suitable assumptions and draw neat figures wherever required.
- Use of non-programmable calculator is allowed.
- Show necessary calculations.

SECTION -I **[20]**

Q - II

[A] Complete the following reaction and comment on the formation of product(s). **[04]**

i)



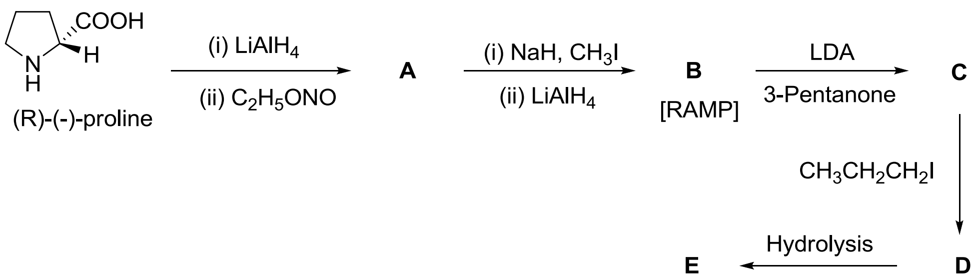
(Use Cram's Rule)

ii)

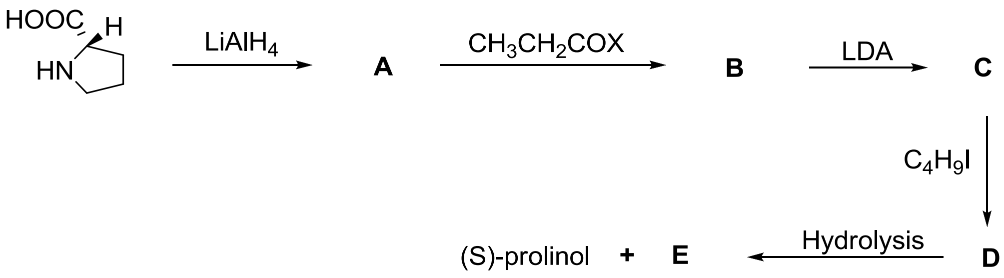
(S)-3-Phenylbutanone $\xrightarrow{\text{(i) LiAlH}_4}$ (?)

(Apply Felkin-Ahn model)

[B] Complete and rewrite the following synthesis. **[03]**



[C] Predict the products **A** to **E**. **[03]**



OR

[C] How one can separate Racemic mixture of 2-butanol with the help of (R, R)-tartaric acid? **[03]**

Q-III

[A] The optical rotation of sample compose of only 2-butanol is determined α_{obs} = 1.0°, $[\alpha]_D$ is known to be 13.52 for (+)-2-butanol. If the path length was 0.6 dm and the concentration of 2-butanol sample was 0.20 g/ml, what is the ee? Calculate the % of (+) and (-) isomer. **[04]**

[B] Give the difference between Multi-mode cavity Microwave Apparatus Vs. Mono-mode cavity Microwave Apparatus. **[03]**

[C] "If two samples containing solvent water and dioxane respectively heated in a single mode microwave cavity at a fixed radiation power and for fixed time, then the final temperature will be higher in water" Explain on the basis of its mechanism. **[03]**

OR

[C] The specific rotation of (+) Mandelic acid is +158°. What would be the observed specific rotation of the following? **[03]**

(i) 40% (-) + 60% (+)

(ii) 50% (-) + 50% (+)

(iii) 75% (-) + 25% (+)

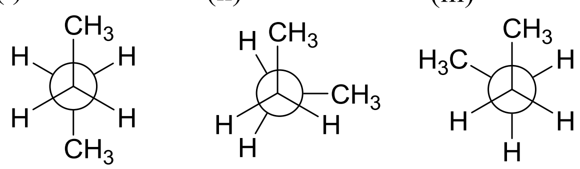
SECTION II **[30]**

Q-IV

[A] Torsional strain for eclipsing H – H bonds has an energy of about 1.0 kcal/mol. Torsional strain for a CH_3 – H eclipse has an energy of about 1.6 kcal/mol. Torsional strain for a CH_3 – CH_3 eclipse has an energy of about 3.5 kcal/mol and torsional strain for a CH_3 – CH_3 gauche has an energy of about 0.9 kcal/mol. Use this information to draw a plot of energy versus dihedral angle for n-butane looking down the C_2– C_3 bond. Calculate and label the difference in energy between the highest energy conformer and the lowest energy conformer. **[04]**

[B] Give the name of following conformations by applying Klyne-Prelog nomenclature. **[03]**

(i) (ii) (iii)



(Staggered) (Eclipsed) (Staggered)

[C] By considering 1,2-dimethylcyclohexane, it contains two stereo-centers. If you calculate the total number of stereoisomer, it was found to be four. Among four possible stereoisomers only three are isolable. Why only three stereoisomers isolable by considering chair form of 1,2-dimethylcyclohexane? **[03]**

Explain.

OR

[C] Write the conformation of cis and trans isomers of 1-bromo-4-phenylcyclohexane and comment on the favored conformation in each case by calculating their relative energies. **[03]**

Given: Energy difference showing preference of equatorial over axial position in mono-substituted cyclohexanes

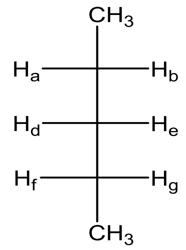
(i) $-C_6H_5$ = 2.9 kcal/mol

(ii) $-Br$ = 0.5 kcal/mol

Q-V

[A] Cycloaddition reaction of butadiene with ethylene proceeds by thermal induction but not by photochemical induction. Justify your answer by giving suitable example. **[04]**

[B] By considering the following compound, Predict the topicity of ……. **[03]**



(i) H_a and H_b (ii) H_d and H_e (iii) H_a and H_d

(iv) H_a and H_f (v) H_b and H_g (vi) H_b and H_e

[C] Discuss heterogeneous phase reaction in ultrasound-assisted organic synthesis. **[03]**

OR

[C] Do as directed. **[03]**

(i) Draw the eclipsed conformation of 2-chloropropanamide. (Look C_2-C_3 bond)

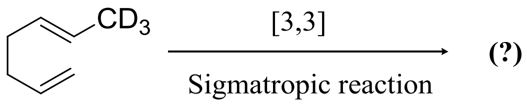
(ii)Draw the staggered conformation of 3-chloropantan-2-ol. (Look C_2-C_3 bond)

(iii)Draw the anti-confirmation of 1,2-dichloro-1,2-difluoroethan. (Look C_1-C_2 bond)

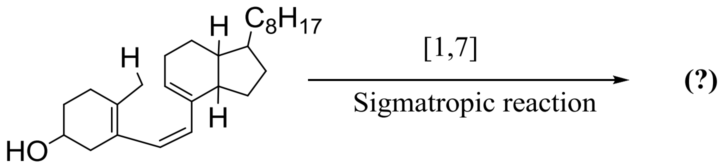
Q-VI

[A] Complete and rewrite the following reactions. **[04]**

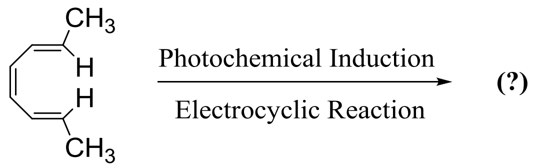
(i)



(ii)



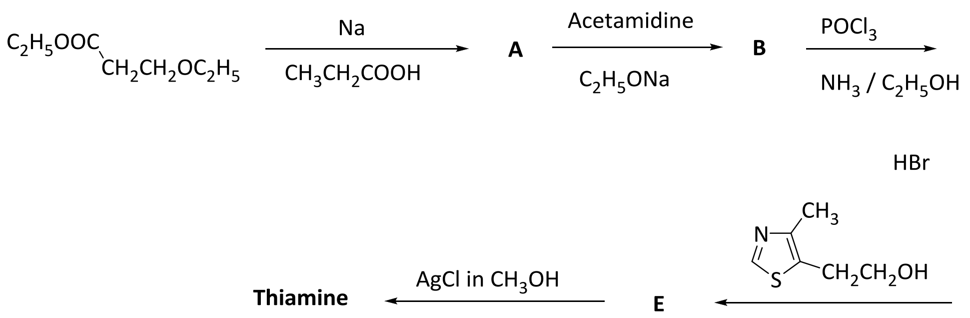
(iii)



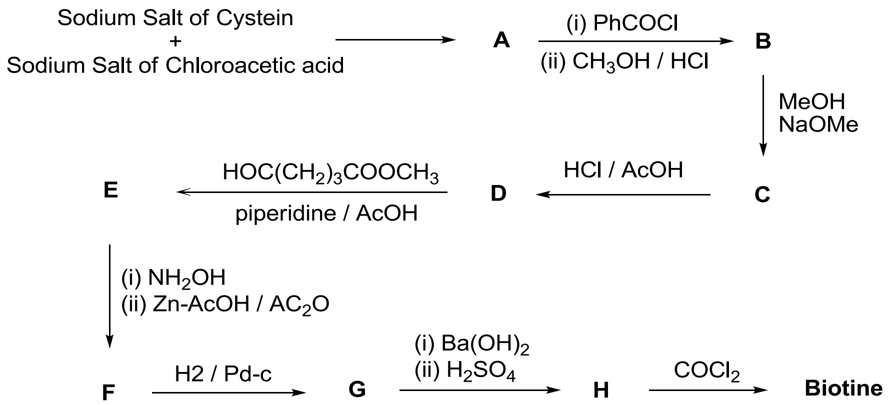
(iv)

$$\text{(2E,4Z,6Z)-Octatriene} \xrightarrow[\text{Electrocyclic Reaction}]{\text{Heat}} (?)$$

[B] Complete and rewrite the following reaction path. **[03]**



[C] Complete and rewrite the following reaction path. **[03]**



OR

[C] "[2+2] cycloaddition reaction proceeds by photochemical induction but not by thermal induction." Justify your answer by giving suitable example. **[03]**